

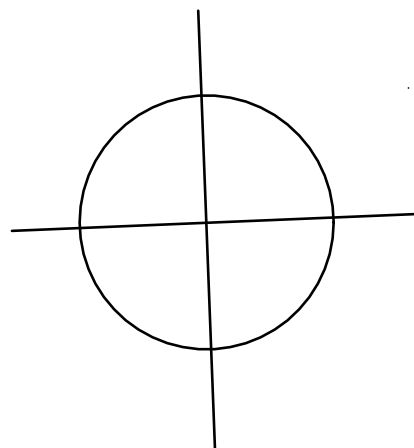
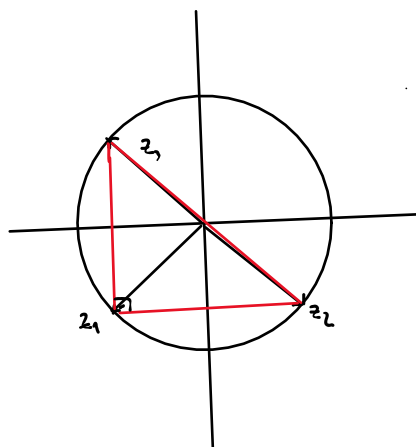
IME - 2021/12

5ª QUESTÃO	Valor: 0,25
Seja $\alpha \in \mathbb{R}$ e z_1, z_2, z_3 números complexos tais que $z_1 = z_2 = z_3 = 4$ e $z_1 \neq z_2$. O menor valor de $\alpha z_1 - (\alpha - 1)z_2 - z_3$, é:	
(A)	$\frac{1}{8} z_1 + z_2 $
(B)	$\frac{1}{4} z_1 - z_2 $
(C)	$\frac{1}{8} z_3 - z_1 z_3 - z_2 $
(D)	$\frac{1}{4} z_1 - z_2 - z_3 $
(E)	$ z_3 $

Problem 1. Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = R$ and $z_2 \neq z_3$. Prove that

$$\min_{a \in \mathbb{R}} |az_2 + (1-a)z_3 - z_1| = \frac{1}{2R} |z_1 - z_2| \cdot |z_1 - z_3|.$$

(Romanian Mathematical Olympiad—Final Round, 1984)



$$z_1, z_2, z_3 \in \mathbb{C}, a \in \mathbb{R} : d(z_1, z_2): \mathbb{C} \times \mathbb{C} \rightarrow [0, \infty)$$

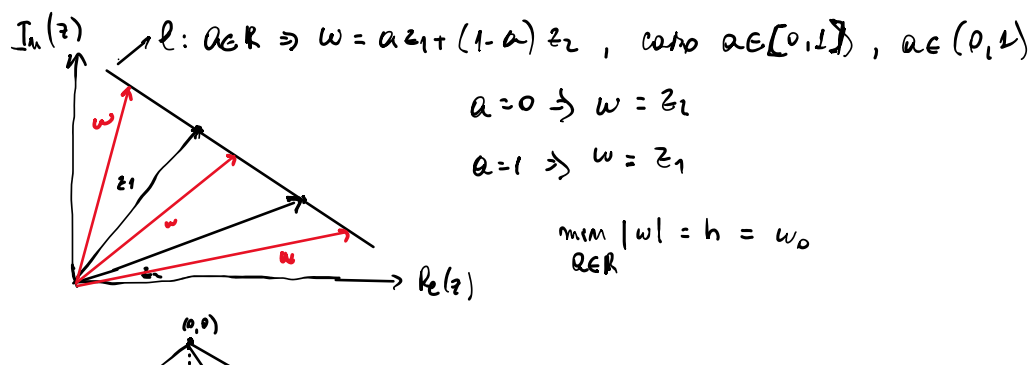
$$d(z_1, z_2) = |z_1 - z_2|$$

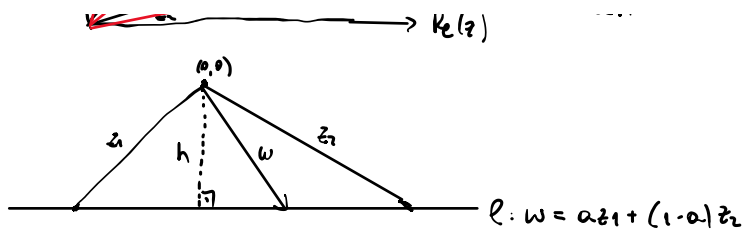
$$i) d(z_1, z_2) = d(z_2, z_1)$$

$$ii) d(z_1, z_2) = 0 \Leftrightarrow z_1 = z_2$$

$$iii) d(z_1, z_2) \leq d(z_1, z_3) + d(z_3, z_2)$$

$$iv) d(z_1, z_2) = d(z_1, z_3) + d(z_3, z_2) \Leftrightarrow (z_1 - z_3) = \lambda (z_3 - z_2)$$





$$\langle z_1, z_2 \rangle : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{R}$$

$$w = a z_1 + (1-a) z_2 = z_2 + a(z_1 - z_2), \quad v = z_1 - z_2, \quad v \in \mathbb{C}$$

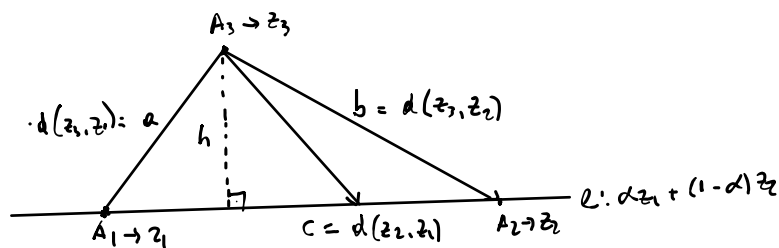
$$w = z_2 + a v$$

$$0 = \langle w, v \rangle = \langle a v + z_2, v \rangle = a \langle v, v \rangle + \langle z_2, v \rangle$$

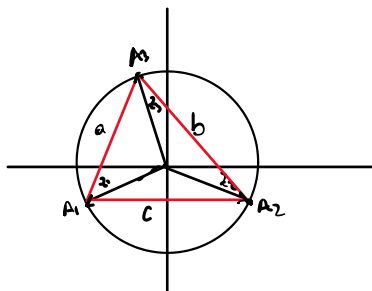
$$a_0 = - \frac{\langle z_2, v \rangle}{\langle v, v \rangle} = - \frac{\langle z_2, v \rangle}{|v|^2}$$

a_0 é valor de $a \in \mathbb{R}$ que minimiza $|w| \Leftrightarrow h = \min_{a \in \mathbb{R}} |w|$

$$w = a z_1 + (1-a) z_2 - z_3$$



$$|z_1| = |z_2| = |z_3| = R$$



$$A_D = \frac{abc}{4R}$$

Δ é inscrito na circ. de raio R
 a, b, c são lados do Δ .

$$A_{A_1 A_2 A_3} = \frac{1}{2} h \cdot c = \frac{abc}{4R} \Rightarrow h = \frac{ab}{2R}$$

$$|z_1| = |z_2| = |z_3| = 4$$

$$h = \min_{a \in \mathbb{R}} |a z_1 + (1-a) z_2 - z_3|$$

$$\left. \begin{aligned} a &= d(z_3, z_1) = |z_1 - z_3| \\ b &= d(z_3, z_2) = |z_3 - z_2| \end{aligned} \right\} \rightarrow h = \frac{1}{8} |z_1 - z_3| \cdot |z_3 - z_2|$$